Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/24/2020
PI Name:	Bryan, Noelle Ph.D.		
Project Title:	Genomic and Functional Analysis of Desiccation Tolerance of International Space Station Isolated Enterococcus faecalis and Their Pathogenicity in Caenorhabditis elegans Genomic and Functional Analysis of Biofilm Morphotypes of International Space Station Isolated Staphylococcus epidermidis and Their Pathogenicity in Caenorhabditis elegans)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nbryan5@mit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Department of Earth, Atmospheric, and Planetary Sciences		
PI Address 2:	77 Massachusetts Ave., 54-918		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	7
Comments:	Dr. Bryan moved to NASA Ames Research Center in December 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-MOBE. Appendix B: Post-Doctoral Fellowships to Study the Microbiome of the ISS as a Built Environment
Start Date:	09/13/2017	**End Date:**	11/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: This was extended to 9/30/2020; however, PI moved to NASA Ames Research Center and the fellowship then ended as of 11/30/2019 (Ed., 2/4/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zuber, Maria Ph.D. (MENTOR/ Massachusetts Institute of Technology) Carr, Christopher Ph.D. (Massachusetts Institute of Technology)		
Grant/Contract No.:	80NSSC17K0688		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by NASA as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. NEW TITLE--July 2018: Genomic and Functional Analysis of Desiccation Tolerance of International Space Station Isolated Enterococcus faecalis and Their Pathogenicity in Caenorhabditis elegans. Change in organism used in this project, approved by NASA Headquarters. Enterococcus faecalis is an opportunistic human pathogen consistently isolated and detected by molecular analysis on board the International Space Station (ISS). The ability of this organism to survive outside of the human gut may have important implications for its spread to a new host environment. The goal of this study is to characterize the desiccation survival capacity of ISS isolated E. faecalis strains and to investigate their pathogenicity in the model organism Caenorhabditis elegans. First, we shall provide the first comparative genomic information of E. faecalis strains to characterize strain-specific sequence variants and the identification of traits that may confer increased resistance to desiccation and antibiotics. We will also examine the presence of the clustered regularly interspaced short palindromic repeats-CRISPR-associated proteins (CRISPR-Cas) that confer bacterial resistance to foreign DNA and may provide an additional method of persistence to the transfer of mobile genetic elements. Phenotypic data will determine how desiccation tolerance varies, and how these variations relate to the clinical and commensal strain types. In vivo pathogenicity tests will culminate our investigation into the effect of E. faecalis biofilm morphotypes on the ability to establish infection in the model organism Caenorhabditis elegans. This work will help to determine genomic, phenotypic, and physiological basis for various desiccation and antibiotic resistance of ISS isolated E. faecalis, and provide insight to pathogenicity of spaceflight isolated microorganisms, where astronaut immunity is decreased and the availability of antibiotics is limited.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will investigate the potential for opportunistic human pathogens, E. faecalis, to survive outside of the human gut. These isolates have not been genetically or phenotypically characterized and those are the first steps of this project.
Task Progress:	Enterococcus faecalis is a multidrug resistant, opportunistic human pathogen and a leading cause of hospital acquired infections. Recently, isolates have been recovered from the air and surfaces onboard the International Space Station (ISS). Pangenomic and functional analyses were carried out to assess their potential impact on astronaut health. Genomes of each ISS isolate, and both clinical and commensal reference strains, were evaluated for their core and unique gene content, acquired antibiotic resistance genes, phage, plasmid content, and virulence traits. In order to determine their potential survival when outside of the human host, isolates were also challenged with three weeks of desiccation at 30% relative humidity. Finally, pathogenicity of the ISS strains was evaluated in the model organism Caenorhabditis elegans. At the culmination of this study, there were no defining signatures that separated known pathogenic strains from the more commensal phenotypes. As a result, the current reliance on database information alone must be shifted to experimentally evaluated characteristics of clinically relevant microorganisms. By harnessing the technology already onboard the ISS, the crew will then be able to achieve near-real time detection of isolates that may impact crew health.
Bibliography Type:	Description: (Last Updated: 02/01/2021)
Articles in Peer-reviewed Journals	Bryan NC, Lebreton F, Gilmore M, Ruykun G, Zuber MT, Carr CE. "Genomic and functional characterization of Enterococcus faecalis isolates recovered from the International Space Station and their potential for pathogenicity." Front Microbiol. 2021 Jan 11;11:515319. https://doi.org/10.3389/fmicb.2020.515319 ; PMID: 33505359; PMCID: PMC7829349 [Note previously reported in January 2020 as Submitted to Frontiers in Microbiology] , Jan-2021
Articles in Peer-reviewed Journals	Ung L, Bispo PJM, Bryan NC, Andre C, Chodosh J, Gilmore MS. "The best of all worlds: Streptococcus pneumoniae conjunctivitis through the lens of community ecology and microbial biogeography." Microorganisms. 2020 Jan;8(1):46. https://doi.org/10.3390/microorganisms8010046 ; PubMed PMID: 31881682 , Jan-2020
Articles in Peer-reviewed Journals	Carr CE, Bryan NC, Saboda K, Mojarro A, Hachey K, Bhattaru SA, Ruykun G, Zuber MT. "Nanopore sequencing at Mars, Europa, and Microgravity conditions." npj Microgravity. 2020 Sep 7;6:24. https://doi.org/10.1038/s41526-020-00113-9 ; PMID: 32964110; PMCID: PMC7477557 [Note previously reported in January 2020 as Submitted to npj Microgravity (NPJMGRAV-00458)] , Sep-2020